Roll No. | | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-7)

INSTRUMENTAL METHODS OF ANALYSIS-THEORY

Subject Code : BP-701T

M.Code : 78387

Date of Examination : 19-05-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. *Write briefly :*

   a. What are chromophores and auxochromes?

   b. What is the Beer-Lambert law?

   c. Define affinity chromatography.

   d. What is fluorescence quenching?

   e. Give the name of two detectors used in IR spectroscopy.

   f. What is flame photometry?

   g. What is atomic absorption spectroscopy?

   h. Define column chromatography.

   i. What is the principle of electrophoresis?

   j. What are the sources of radiation used in U-V-Visible spectroscopy?

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SECTION-B

2. Explain the theory and instrumentation of UV-Visible spectroscopy. Discuss the factors affecting the absorption spectra.

3. Explain the theory and instrumentation of fluorimetry. Discuss the factors affecting fluorescence.

4. Explain the theory, instrumentation and applications of high-performance liquid chromatography (HPLC).

SECTION-C

5. What are spectral shifts? Discuss the solvent effect on absorption spectra.

6. Write a note on nepheloturbidometry?

7. What are the sources of radiation, wavelength selectors, and detectors used in Flame Photometry?

8. Explain the concept of electronic transitions and chromophores in UV-visible spectroscopy.

9. What is the instrumentation used in affinity chromatography?

10. What are the detectors used in UV-Visible spectroscopy?

11. Discuss the factors affecting electrophoretic mobility in Electrophoresis.

12. What are the properties of ion exchange resins?

13. What is the mechanism of ion exchange process in ion exchange chromatography?

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student

-2023